

# MATHIAS VERNIER

## EMBEDDED SYSTEMS ENGINEER

### EDUCATION

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**ECE Paris - Graduate School of Engineering** 2023 – 2028

*Master's Degree in Engineering – Major in Embedded Systems (ING4)*

- **Relevant Coursework:** Real-Time OS, Computer Architecture, Signal Processing, Hardware AI, Control Theory, Embedded Systems Design, Digital Logic & RTL.

### EXPERIENCE

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**Engineering Intern – IMOCA Digital Twin** Jan 2026 – Feb 2026

*4CAD Group – Vendée Globe 2028 Project*

- Optimized native CAD data to build a real-time 3D VR cockpit validation tool (60+ FPS) for skipper Benjamin Dutreux.
- Engineered a headset-free cockpit walkthrough using motion tracking sensors, deployed live at Global Industrie 2026.
- Created sensor-driven virtual environments and photorealistic assets with KeyShot for pre-manufacturing reviews.

**Engineering Intern – Automated Workflows** Jan 2025 – Feb 2025

*Groupe SYD*

- Architected automated data pipelines and event-driven workflows using Power Platform, reducing manual error rates.
- Designed backend business logic, validation rules, and automated testing routines to ensure end-to-end data integrity.

### TECHNICAL PROJECTS

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**H2-Vision – Fuel Cell Telemetry Unit** In Progress

*ESP32 · KiCad (PCB) · FreeRTOS · INA226 · SPI/I2C · Python · H2GP*

- Designing a  $\leq 30$  g PCB (KiCad, INA226) to monitor H2 fuel cell power, temperature, and vehicle speed.
- Developing dual-core FreeRTOS firmware on ESP32 (50 Hz sampling, live telemetry to Python GUI) for H2GP teams.

**Fruit Ninja on FPGA – Pure Hardware Logic** Academic Project

*Intel MAX 10 (DE10-Lite) · VHDL · Quartus Prime · VGA DAC · RTL*

- Engineered an arcade game in pure RTL VHDL with synchronous 25 MHz pipelines on Intel MAX 10 FPGA and VGA DAC.
- Implemented real-time physics for 13 sprites and blade control via HC-SR04 ultrasonic sensing and rotary encoder.

**Drawbot – Autonomous Plotter Robot** Academic Project

*ESP32 · C++ · PID Control · Kinematics · Logic Analyzer · Wi-Fi*

- Implemented dual closed-loop PID controllers (linear/angular) in C++ on ESP32 with pen-offset differential kinematics.
- Embedded an asynchronous Wi-Fi server for real-time control, debugging PWM timings with oscilloscopes and analyzers.

### SUMMARY

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Hands-on Embedded Systems Engineer bridging custom board design, deterministic real-time firmware, and pure RTL logic. Proven experience architecting low-overhead electronic systems from KiCad schematics and FreeRTOS tasks down to cycle-accurate FPGA pipelines. Seeking a 5-month international internship starting April 2027.

### CONTACT

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mathias.vernier@outlook.fr

+33 6 78 44 10 18

Paris, France

[mathiascece.github.io](https://mathiascece.github.io)

### TECHNICAL SKILLS

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#### Programming

C, C++, VHDL, Python

#### Hardware & Embedded

ESP32 (Dual-Core), FreeRTOS, KiCad (PCB Design), Intel MAX 10 FPGA, STM32, ARM Cortex, INA226, I2C, SPI, PID Control

#### Tools & Instrumentation

Logic Analyzer, Oscilloscope, Multimeter, Precision Shunts, Quartus Prime, Jupyter, Git, Linux

### LANGUAGES

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**French:** Native

**English:** Advanced (TOEIC 870)

### EXTRACURRICULAR

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#### Paris Fire Brigade (BSPP)

*One-Day Crisis Simulation Exercise*

- Served as an emergency responder in a full-scale counter-terrorism drill.
- Applied rapid triage protocols and crisis communication under high stress.